

Theory 4. Conjugate functions and norms, proximal operators

Task 1 (1.5 pts) For each of the following functions f find its Fenchel conjugate function f^* :

- a) $f : [0, +\infty) \rightarrow \mathbb{R}$, $f(x) = x \log(x) - x$, if $x > 0$, and $f(x) = 0$, if $x = 0$;
- b) $f : (0, +\infty) \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x}$;
- c) $f : (-\infty, 0] \rightarrow \mathbb{R}$, $f(x) = -2\sqrt{-x}$;
- d) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \max(0, x)$.

Task 2 (2 pts) For each of the following functions f find its Fenchel conjugate function f^* :

- a) $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f(\mathbf{x}) = \log(\sum_{i=1}^n \exp(x_i))$;
- b) $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f(\mathbf{x}) = \max(x_1, \dots, x_n)$;
- c) $f : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$, $f(X) = -\log \det(X)$.

Task 3 (1.5 pts) For each of the following functions find its subdifferential:

- a) $f(\mathbf{x}) = \|\mathbf{x}\|_\infty$;
- b) $f(\mathbf{x}) = \sum_{1 \leq i < j \leq n} |x_i - x_j|$.

Task 4 (1.5 pts) The spectral norm of a matrix is defined as its maximal singular value: $\|A\|_2 = \sigma_{\max}(A)$. The matrix conjugate norm is defined as $\|A\|_{2*} = \max_{X: \|X\|_2=1} \langle A, X \rangle$. Show that for spectral norm

- a) the maximization problem in conjugate norm can be transformed to the optimization w.r.t diagonal matrices;
- b) derive from the previous item that $\|A\|_{2*} = \sum_{i=1}^n \sigma_i(A)$.

Task 5 (1.5 pts) Suppose that $f : E \rightarrow \mathbb{R}$ is closed and convex function. Show that the following conditions are equivalent:

- a) $u \in \text{prox}_f(x)$;
- b) $x - u \in \partial f(u)$;
- c) $f(v) \geq f(u) + \langle x - u, v - u \rangle$.

Task 6 (2 pts) Suppose that $f : E \rightarrow \mathbb{R}$ is closed and convex function. Show that

$$\|\text{prox}_f(x) - \text{prox}_f(y)\|_2^2 \leq \langle \text{prox}(x) - \text{prox}(y), x - y \rangle \quad \forall x, y.$$

Derive from this result, that proximal operator is a contraction:

$$\|\text{prox}_f(x) - \text{prox}_f(y)\| \leq \|x - y\| \quad \forall x, y.$$